Does Underlying Structure Predict Quantum Trainability and Dequantization?

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1. Research question

Can the underlying structure of a learning problem predict, before training, whether structure-preserving quantum models of it will be trainable and resist dequantization? Random, structure-free quantum circuits are untrainable at scale because their gradients vanish exponentially with the number of qubits (McClean et al. 2018). Building a problem's symmetry into the circuit keeps the gradients large (Larocca et al. 2022; Ragone et al. 2024), but the same restriction can let a classical computer imitate the model, so that it offers no quantum advantage (Cerezo et al. 2025). Whether a given problem lands on the useful side is today known only after a model has been built and run. We ask whether it can be read off the problem itself, from four kinds of structure: symmetry, locality, spectrum and data complexity. Answering it needs two ingredients, a large and varied collection of real learning problems, and a way to extract their structure; this deliverable builds the first and tests the second.

2. Data collection

The unit of observation is a *problem*: one binary learning task on n sites (4 to 12, the number of qubits a model of it would use), with its labelled inputs, a fixed 70/30 TRAIN/TEST split and its label rule. Each input x is stored with an operator $M(x)$ that defines a differential equation, and the operator fixes the problem's type (Table 1). The type depends on the mathematics, not the domain: a MaxCut instance encodes like a graph, while a 3-SAT formula has three-site terms a graph cannot have.

Table 1: Types, sources and counts. Groups are independent sources (one dataset, network or generator).

Type	Operator $M(x)$	Sources and collection	Problems	Groups	Samples
GRAPH	graph Laplacian	20 real networks (Netzschleuder API: biology, social, infrastructure, technology), cut into connected pieces of n nodes; real molecules (Alchemy, ZINC via TUDataset)	345	22	100 782
COVARIANCE	precision matrix	EEG motor imagery (MOABB API: BNCI2014_001, Cho2017), one problem per subject and session; 9 sensor datasets (UEA archive)	404	11	87 992
DIAGONAL	clause-penalty Hamiltonian	SAT formulas from 11 standard generators (random k-SAT, XOR, NAE, planted, community, SR)	358	11	107 400
QUANTUM	Hamiltonian, ground state	8 spin-chain models (exact ground states on both sides of a phase transition); 6 molecules (Hartree–Fock, Jordan–Wigner)	261	14	41 756
Total			1 368	58	337 930

Each source was listed, screened, downloaded (with a file hash) and converted by a logged pipeline. A candidate problem was kept only if it had at least 100 samples, at least 50 distinct inputs and at most 20% of equal inputs carrying different labels; every rejection is recorded with its reason (for the real networks, 892 candidates were checked and 307 kept). Labels are either given by the source (which hand was imagined, which class a recording belongs to) or computed exactly from the input (a graph property, the side of a phase transition); when a source gives a score, the label is $+1$ above the TRAIN median, so the TEST data never influence the cut. Real inputs make up 749 problems (55%), collected through the Netzschleuder, TUDataset and MOABB APIs and the UEA archive; the SAT and spin-chain problems are generated from standard models, because in these two domains generated instances are how problems are produced, and every problem carries its origin.

3. Quantitative description and variables

The final dataset holds 1 368 problems, 337 930 samples and 58 groups, with n from 4 to 12 and 1 107 distinct input sets (problems sharing their inputs and differing only in their labels). Class balance lies in $[0.2, 0.8]$ in every problem. Problems from one group are not independent, so the group, not the problem, is the unit of evidence in every later analysis. The variables are listed in Table 2: the first are recorded per problem; the second are computed from the *factor graph* of each operator, a bipartite graph with one node per site and one per term, joined when the term acts on the site, and summarised over a problem's inputs by the median and interquartile range. All are unchanged when the operator is multiplied by a constant.

4. Preprocessing and choices

The raw harvest (1 379 problems) is never modified; a script with six stated rules builds the final dataset from it, and the raw files' checksum is identical before and after. An audit found 0 structural failures and five problems in the content, each handled by one choice. Identical inputs on both sides of the split, in 22 problems, would leak TEST into TRAIN: these problems were re-split with identical inputs kept on one side, rather than deduplicated, which would have changed the population, and repeated inputs are weighted so that each counts once in TEST. One problem below the class-balance

Table 2: Variables.

Variable	Meaning
type, n, samples, split	operator type, number of sites, number of labelled inputs, 70/30 TRAIN/TEST
class balance, label rule	share of $+1$ labels; how the labels were made
distinct inputs, conflict rate	number of different inputs; share of equal inputs with different labels
source, group, origin	where the problem comes from, its independent source, real or generated
equation check	check method, worst residual, number of failures (0 in all 1 368 problems)
terms per site; term sizes	interactions per site; whether terms are pairwise or many-body
site degree spread; support density	whether a few sites dominate; how far interactions reach
symmetry classes and fraction	how many sites look alike under colour refinement (1: all alike, 0: none)
spectral entropy; participation ratio	how many Laplacian modes of the factor graph carry its weight
algebraic connectivity	how hard the factor graph is to cut in two

floor was excluded. In 24% of spin-chain samples the ground state was degenerate, so the stored state was arbitrary and the input not reproducible: the spin chains were re-harvested with ground states fixed inside a symmetry sector, which cut such samples from 642 to 10 on a test run. Two molecule geometries whose Hartree–Fock calculation did not converge had untrustworthy labels and were removed. Finally, 475 sensor samples whose operator collapses to the identity carry no information and act as a label shortcut; they are flagged rather than removed, and every result is reported with and without them. The final dataset passes the audit with 0 failures and 0 leaks. One known gap remains: 10 periodic XY spin-chain problems were lost to a solver failure, since fixed, and are to be re-harvested.